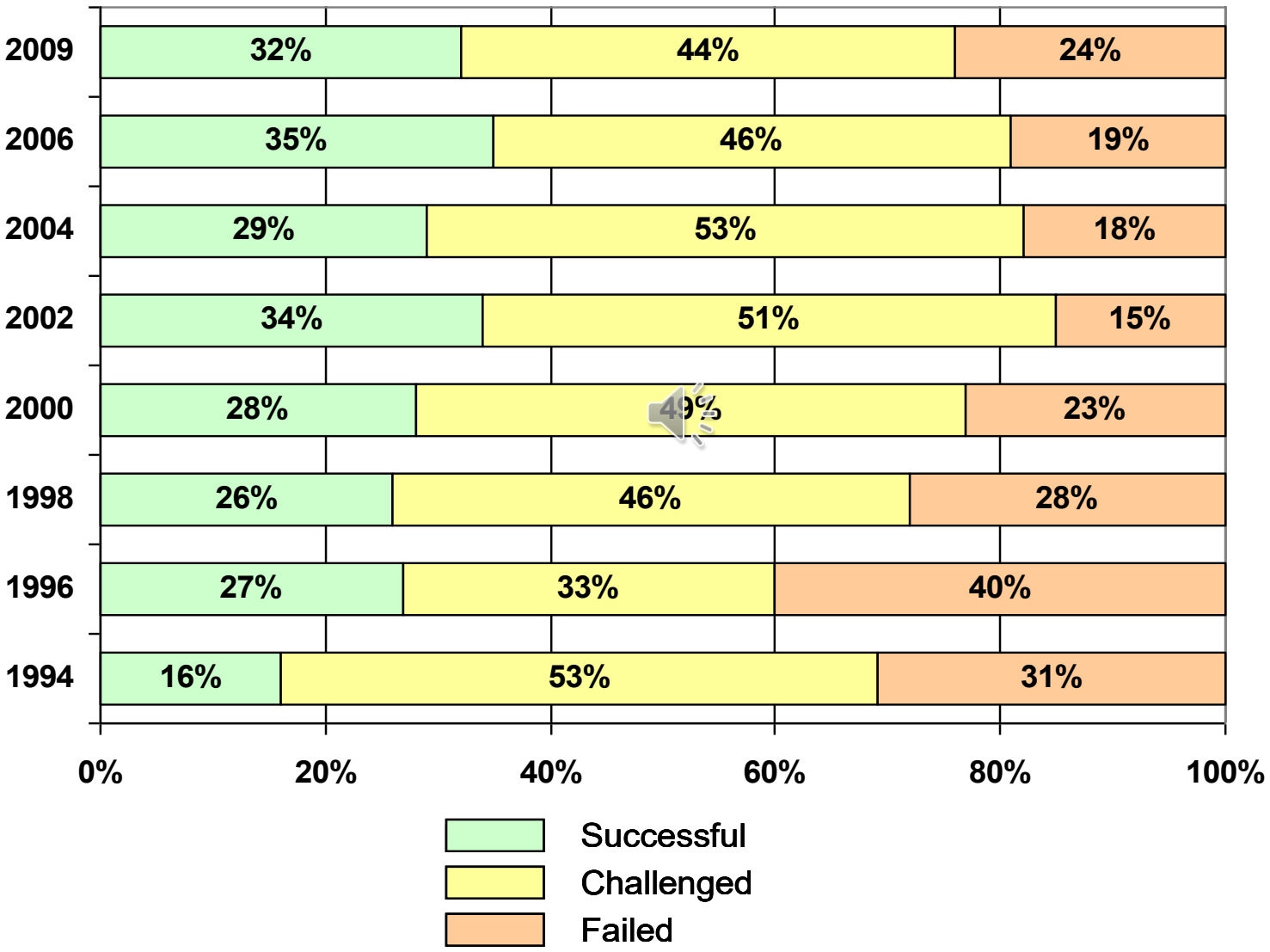
Software Process Management

Lecture 12

Introduction

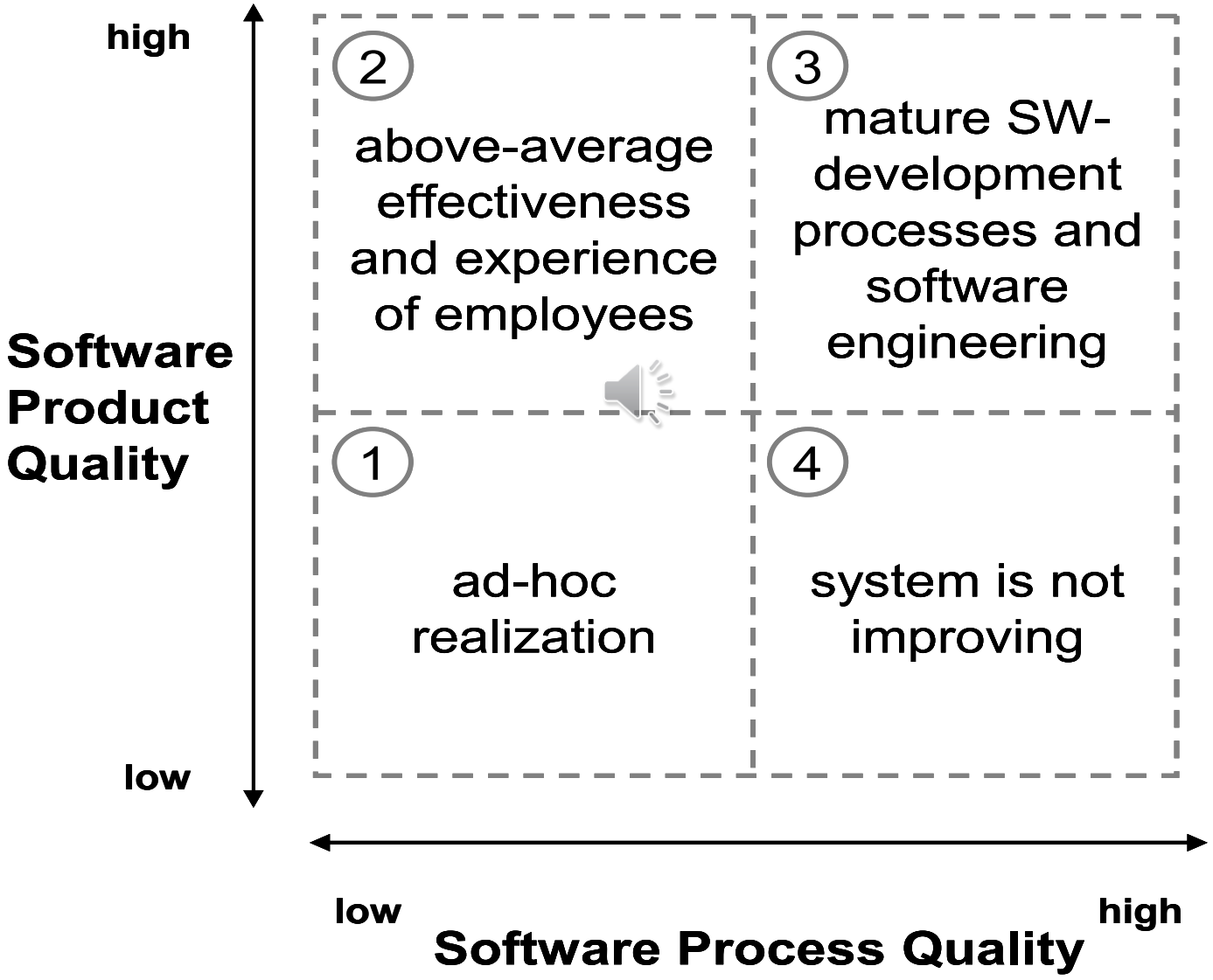
- The track record of software development projects in general is fairly poor according to certain, somewhat debated, surveys
- A survey from industry shows that in the early 1990's, less than a fifth of the projects succeeded well in terms of budget and time used to complete the projects
- Recently, projects have had a higher success rate but still a considerable amount can be considered to have failed



Standish Group, The Chaos Report, 1994-2009

The need for improvement

- The results of the industry surveys indicate that there is room for improvement with respect to the software processes of organizations
- When software processes improve, it is likely that the quality of products improve at the same time
- Achieving high product quality is more difficult if the process is not well tuned - success requires much individual effort in such conditions


high
low
Software Product Quality
2
above-average effectiveness and experience of employees
3
mature SW-development processes and software engineering
1
ad-hoc realization
4
system is not improving
low
high
Software Process Quality

Methods of Improvement

- There are several ways processes can be improved: improvement can for instance be classified as
  - ***model-based*** if process improvement is based on existing reference models or
  - ***continuous*** if improvement is more evolutionary and focused on a set of issues identified in the organization

- Improvement efforts can also be directed by the use of
  - **business alignment methodologies** that point out the key areas where improvement should be done as defined by the overall strategy of the organization

Model Based Improvement

- Model-based improvement involves comparing the various processes found in a specific organization – i.e. how work is carried out - against a reference model
- The predefined reference models include the best practices of the software industry as seen by those that have created the reference model
- There are many kinds of standardized reference models available such as Capability Maturity Model Integration (CMMI) and ISO/IEC 15504 (SPICE)
- Organizations gain knowledge and insight into improvement activities when their processes are assessed
- Assessment and the subsequent improvement activities help to spread organization-wide initiatives as people from various positions participate in process assessment procedures
- External experts are often involved in assessing the practices of an organization and can review the activities objectively

- The organization's capability to fulfill objectives of the reference model can defined as a result of assessment and the attained level of maturity can be used to describe what kind of processes are in place in the organization
- Because of the generic nature of the reference models, there might not be a good fit between a reference model and the actual needs of an organization – following the reference model guidelines might not lead to success
- The review work of external professionals comes at a price and the cost of assessment can be too high for small and medium sized enterprises which makes the improvement approach less popular

An Example - CMMI

- Capability Maturity Model Integration (CMMI) is a reference model that has been designed at the Carnegie Mellon University
- The model is periodically updated: the current version 1.3 was published in 2010 but earlier versions are still in use, especially the previous 1.2 version
- There are three different source models in the current version of CMMI: *development* (CMMI-DEV), *acquisition* (CMMI-ACQ) and *service* (CMMI-SVC)
- A cluster of related practices is called a process area
- The model has been divided into 22 process areas which are further split to four groups of *project management*, *engineering*, *process management* and *support*

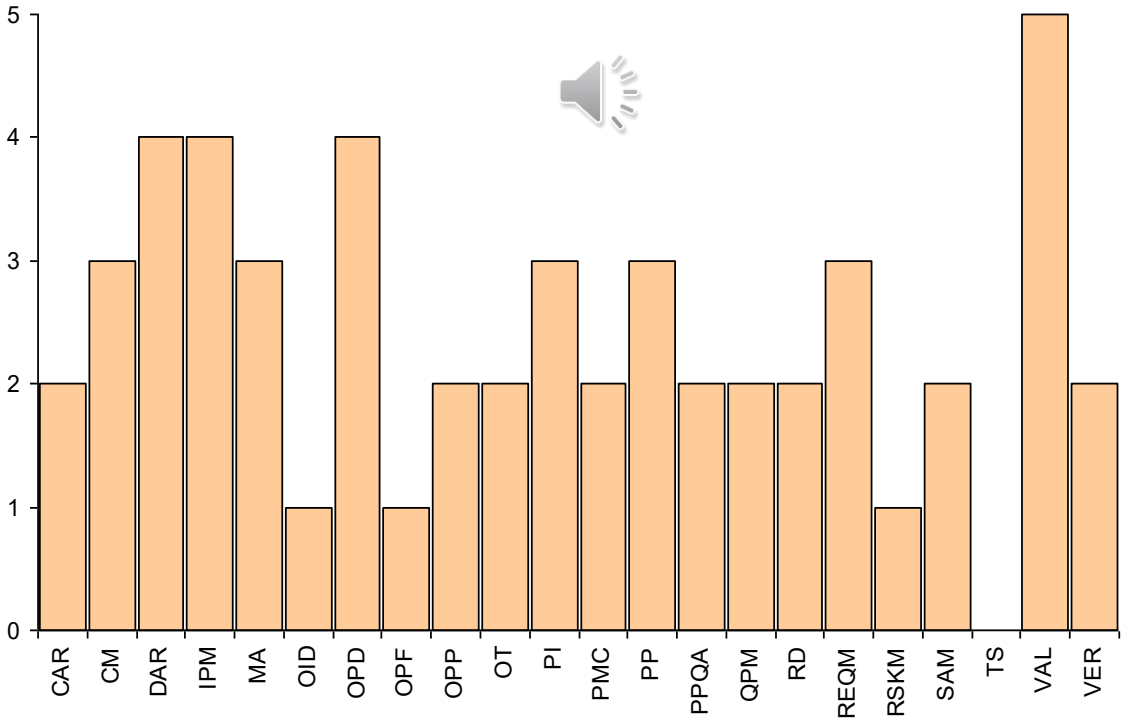
- As a result of the assessment, organizations can be assigned a score which indicates how well the activities in the organization are managed
- *Staged representation* models the state of the organization's processes through one *maturity level* instead of multiple capability levels
- When process areas are evaluated individually, *capability levels* from zero to five can be used for each area and the result can be illustrated in *continuous representation*

Capability Levels (CL)

CL 0 – Incomplete, no requirements
CL 1 – Performed, goals achieved
CL 2 – Managed, managed processes
CL 3 – Defined, standard processes
CL 4 – Quantitatively Managed, statistical control
CL 5 – Optimizing, continuous improvement

Maturity Levels (ML)

ML 1 – Initial, no requirements
ML 2 – Managed, managed processes
ML 3 – Defined, standard processes
ML 4 – Quantitatively Managed, statistical control
ML 5 – Optimizing, continuous improvement



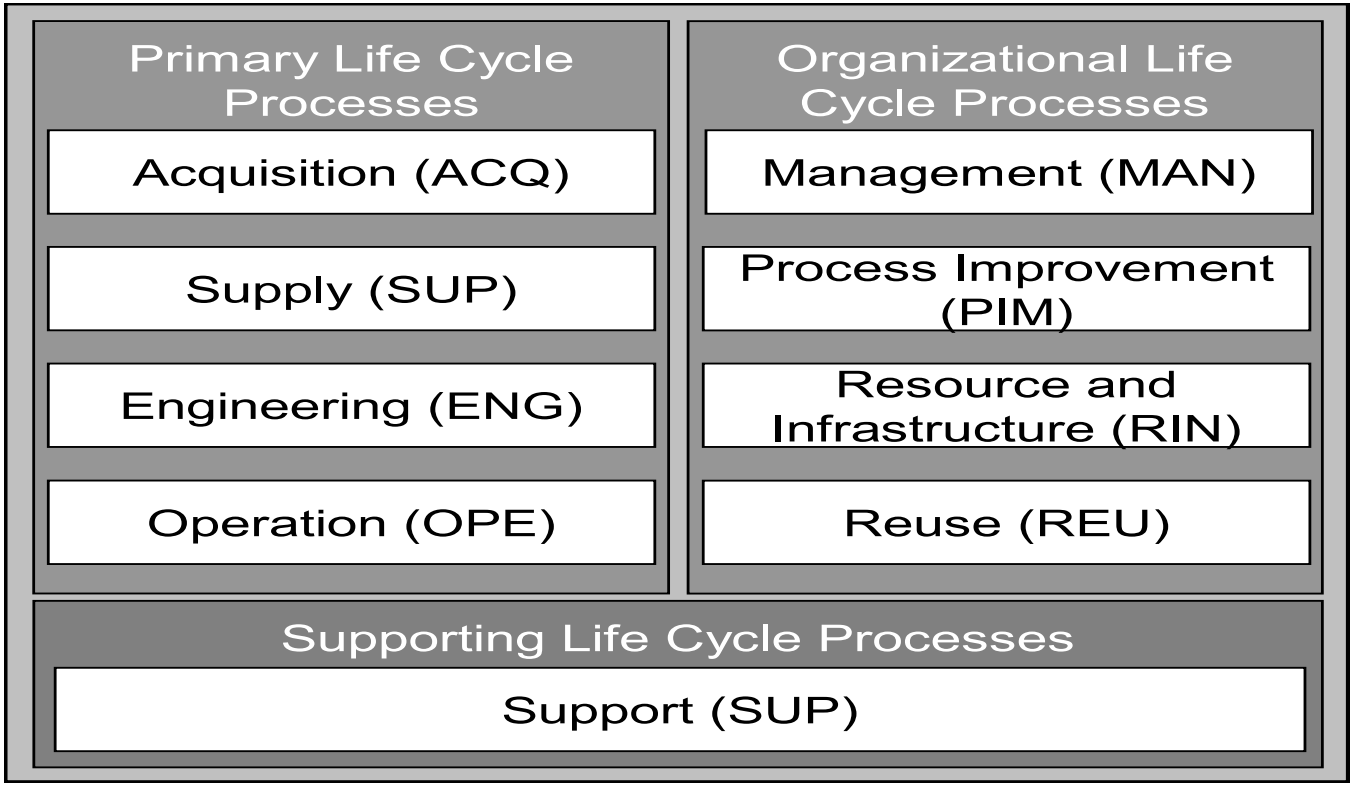
Continuous representation uses capability levels to evaluate individual process areas

ISO/IEC 15504 (SPICE)

- An international five-part standard also known as Software Process Improvement and Capability Determination (SPICE)
- Part five of the standard defines the reference model and states how the areas of the model should be assessed
- The processes in the reference model are divided into three top-level categories: primary-, organizational-, and supporting life cycle processes
- There are a total of 49 processes dispersed in the different categories and each process can be evaluated with the help of process attributes that are defined in the model
- Evaluations in organizations result in assigning a capability level to the process in question, a practice which is akin to continuous representation of CMMI

Capability Levels (CL)

CL 0 – Incomplete, no requirements
CL 1 – Performed, the processes exist
CL 2 – Managed, managed processes
CL 3 – Established, defined processes
CL 4 – Predictable, defined limits
CL 5 – Optimizing, continuous improvement

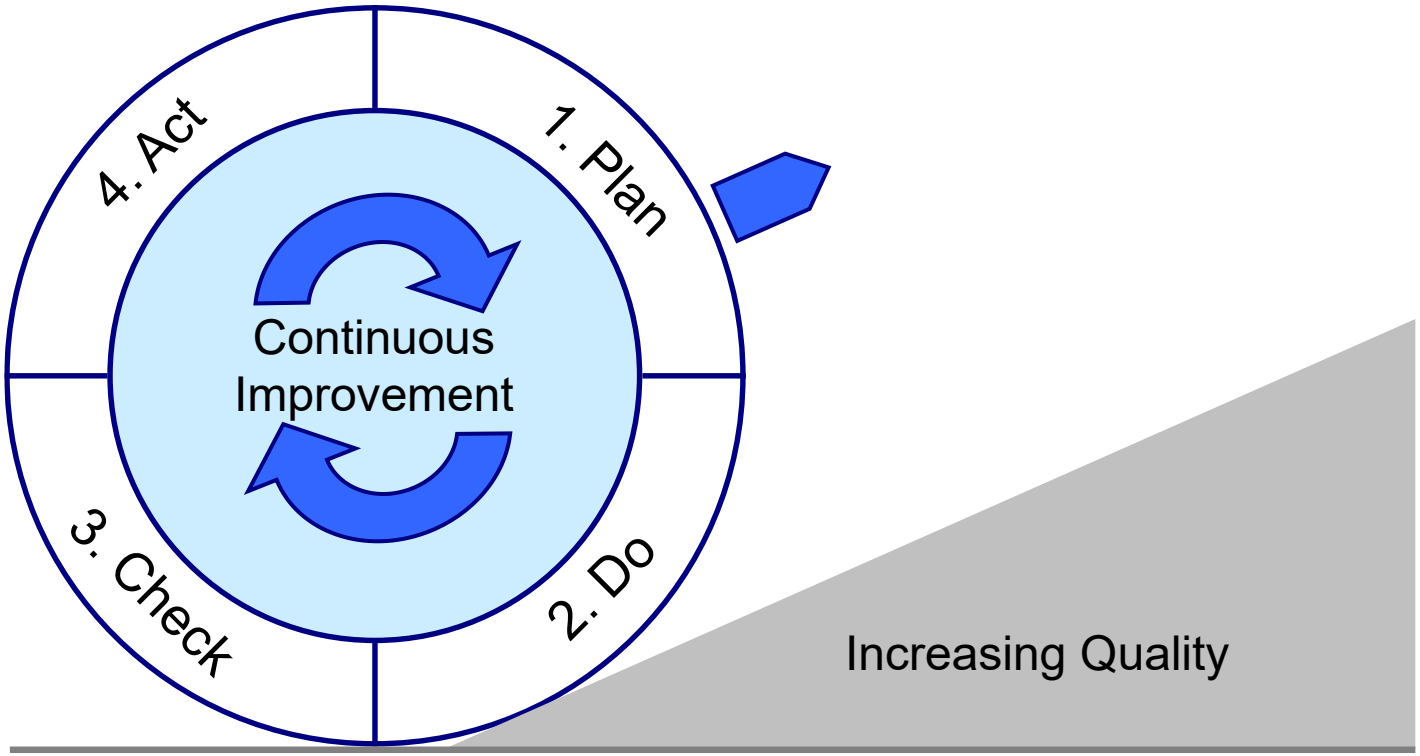


The capability of organizations in SPICE is assessed by reviewing process in three process areas

Continuous process improvement

- Continuous improvement approaches are often problem-based which means that a specific issue is first identified and analyzed after which a solution can be proposed, implemented in a proper fashion and finally the outcome can be reviewed
- On the bright side, this way improvement is *focused* to the actual problems faced by the organization and as such quite *specific* to the organizations; with proper measurement in place, impact of improvement initiatives can be seen without much latency
- Focused improvement activities might not always be a blessing for the organization in that sense that improvement is somewhat local

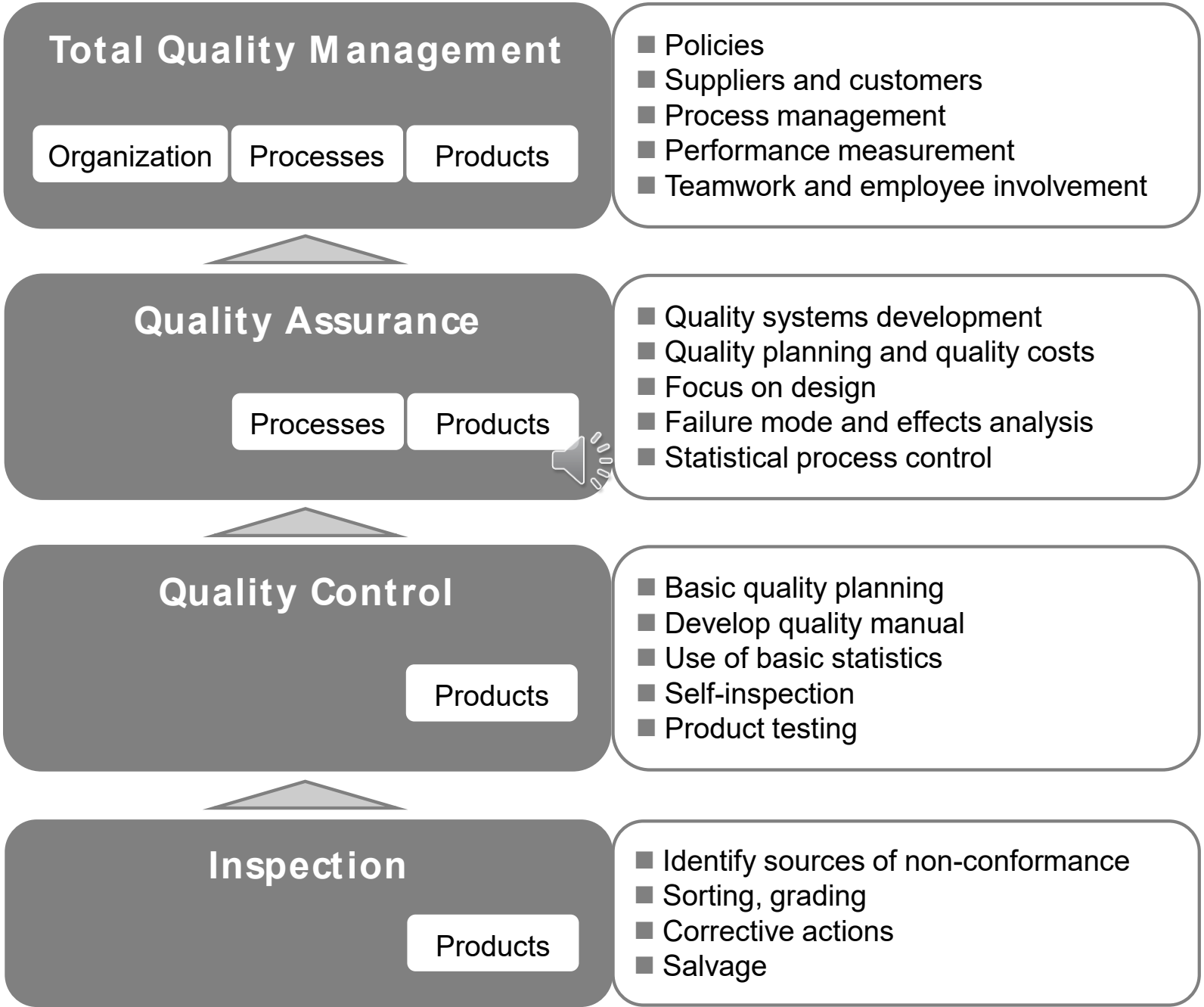
- Plan-Do-Check-Act (PDCA) is a continuous improvement approach that can be used to solve specific problems
- **Plan** - Perform a problem or potential analysis
- **Do** - Implement and perform improvement actions
- **Check** - Analyze the success of the improvement actions
- **Act** - Analyze the differences between the actual and the expected results, determine their causes, and define appropriate means to achieve the expected results


4. Act
1. Plan
Continuous
Improvement
3. Check
2. Do
Increasing Quality

Total Quality Management

- The theory of Total Quality Management originated from the field of industrial process management in United States

- According to the concept of holistic quality, everyone who is involved with a particular product is and should be responsible of both the quality of the product and the quality of the process used to create the product


Total Quality Management
Organization
Processes
Products
Policies
Suppliers and customers
Process management
Performance measurement
Teamwork and employee involvement
Quality Assurance
Processes
Products
Quality systems development
Quality planning and quality costs
Focus on design
Failure mode and effects analysis
Statistical process control
Quality Control
Products
Basic quality planning
Develop quality manual
Use of basic statistics
Self-inspection
Product testing
Inspection
Products
Identify sources of non-conformance
Sorting, grading
Corrective actions
Salvage

Kaizen

- Literally means improvement in Japanese
- Originally the belief in the Japanese industry was that organizations would better achieve their goals if employees were more involved in improving activities in an organization
- The solution was to introduce small, self-managed employee groups that have regular meeting to discuss what could be improved
- Meetings allow groups to set targets for groups
- Employees are potentially more committed to work when they get to present ideas and implement changes
- Continuous improvement and improvement programs mean that employees get training and acquire new skills when implementing new ideas
- Kaizen gives room for leadership and the possibility for employees to step up

Zero Defects Program

- Defects that remain in a product after quality assurance activities lead to increased work later and can be a source of customer dissatisfaction
- When possible, it is better to prevent the introduction of defects rather than only focusing on the correction of defects and this is one of the core ideas of the Zero Defect Program
- The Zero Defect Program can be seen to consist of four principles

1) Quality is defined as conformance to requirements.
2) Defect prevention is preferable to correction.
3) Zero defects is the quality standard.
4) Quality is measured as a price of nonconformance.

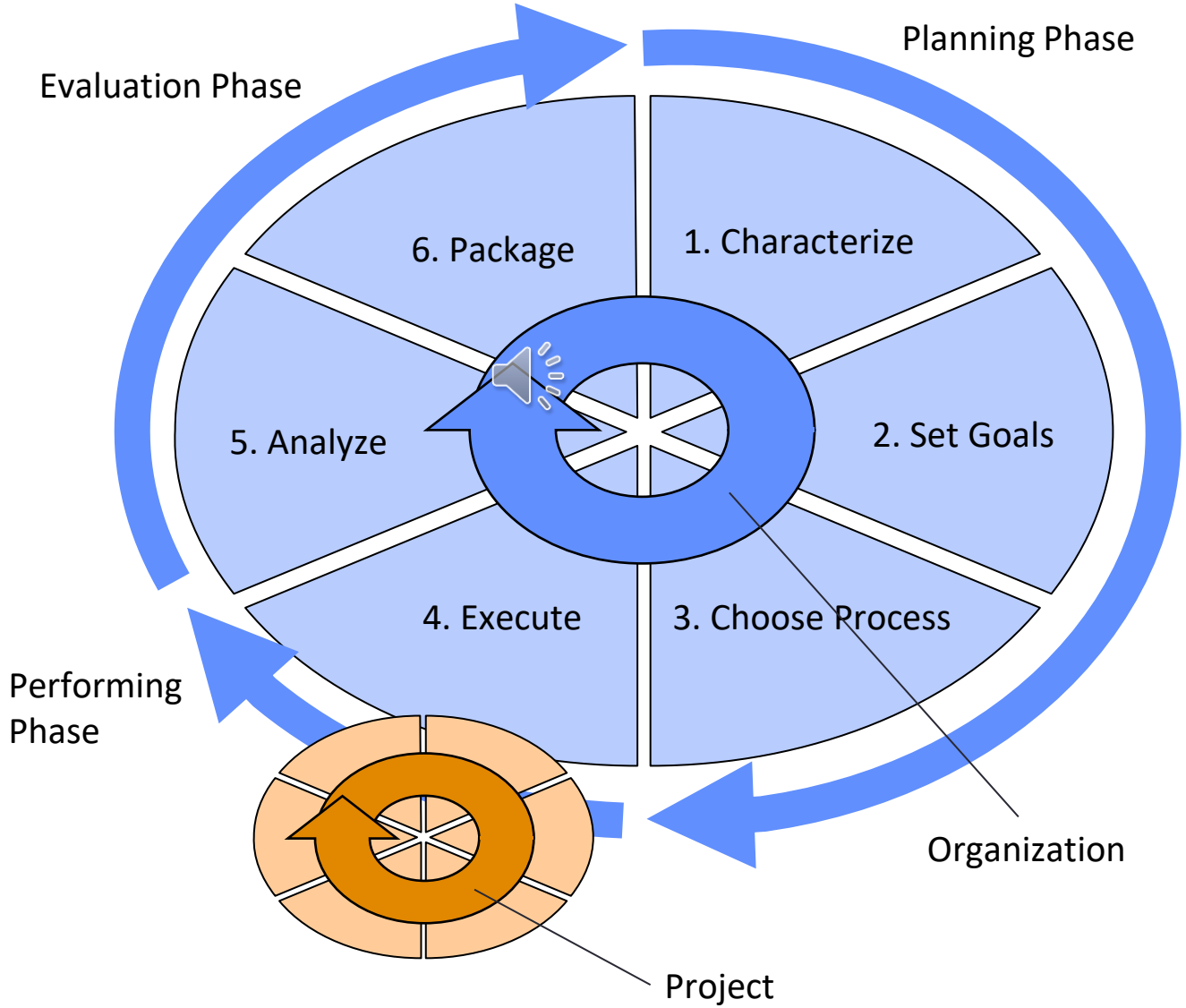
Six Sigma

- Six Sigma (6σ) is a set of practices designed to improve manufacturing and business processes and seeks to identify and eliminate the causes of defects
- Six Sigma was first defined by Bill Smith at Motorola in 1986
- Six Sigma can be used for creating new or controlling existing manufacturing or business processes in order to deliver products or services that meet customer needs
- Six Sigma asserts that continuous efforts to achieve stable and predictable process results (i.e., to reduce process variation) are of vital importance to business success
- Six Sigma adapts the Plan-Do-Check-Act paradigm for continuous improvement with two models
  - DMAIC – define, measure, analyse, improve, control - for improving an existing business process
  - DMADV – define, measure, analyse, design, verify - for creating new product or process designs

Strengths and Weaknesses of 6σ

- **Strengths**
- Utilizes tools and techniques for fixing problems in manufacturing and business processes in a continuous and disciplined fashion
- Focuses on decision making based on facts and data rather than on assumptions
- **Weaknesses**
- Requires large amounts of reliable measurement data, which, in contrast to manufacturing, are typically not available in the software development domain
- Lack of clear connection to business objectives; the sigma-level (defect rate) is assumed to be the major factor contributing to the fulfilment of customer needs

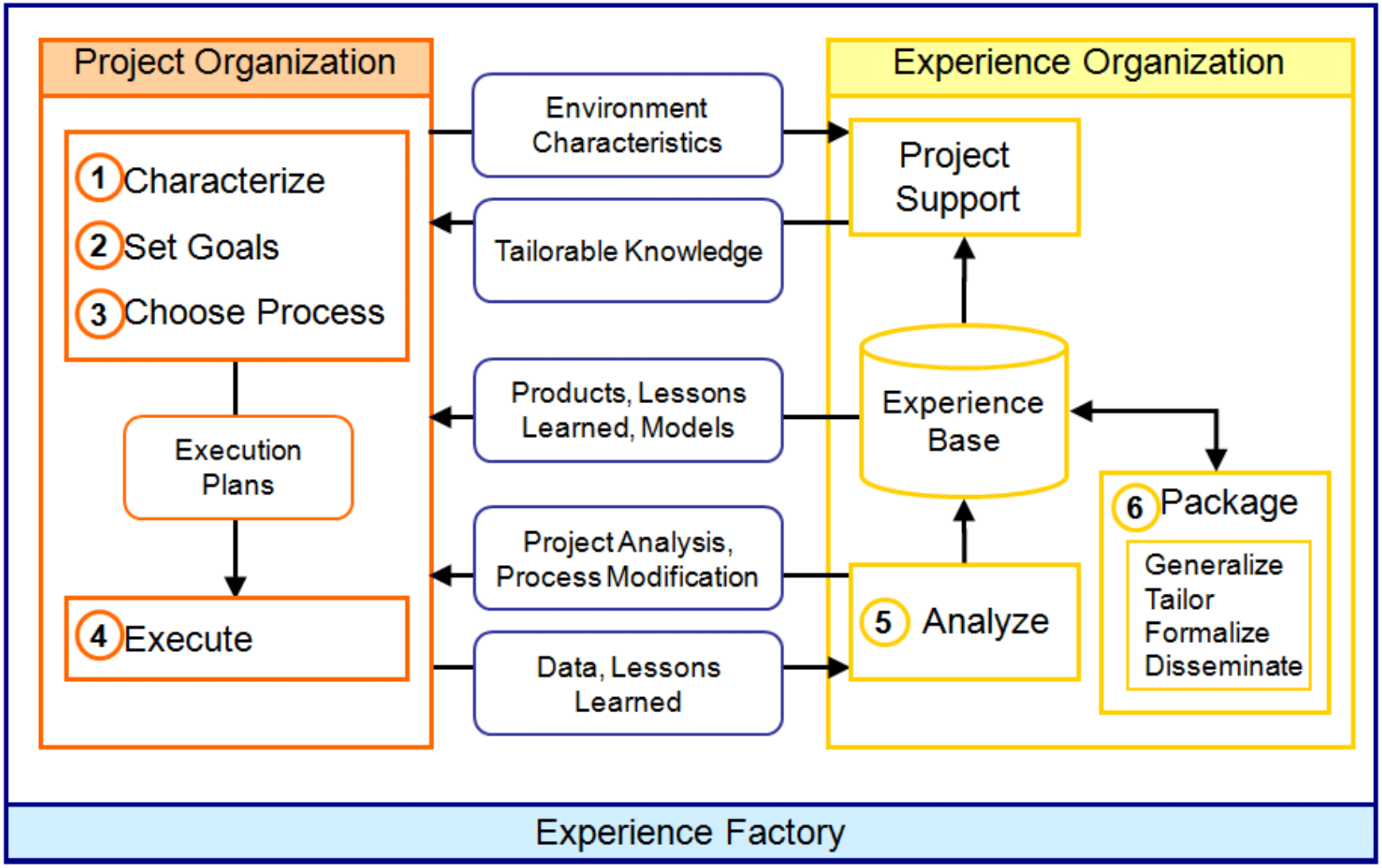
Quality Improvement Paradigm (QIP)



The Experience Factory

- An organization gains experience when it works on projects but knowledge might be lost if it is not properly stored for later use

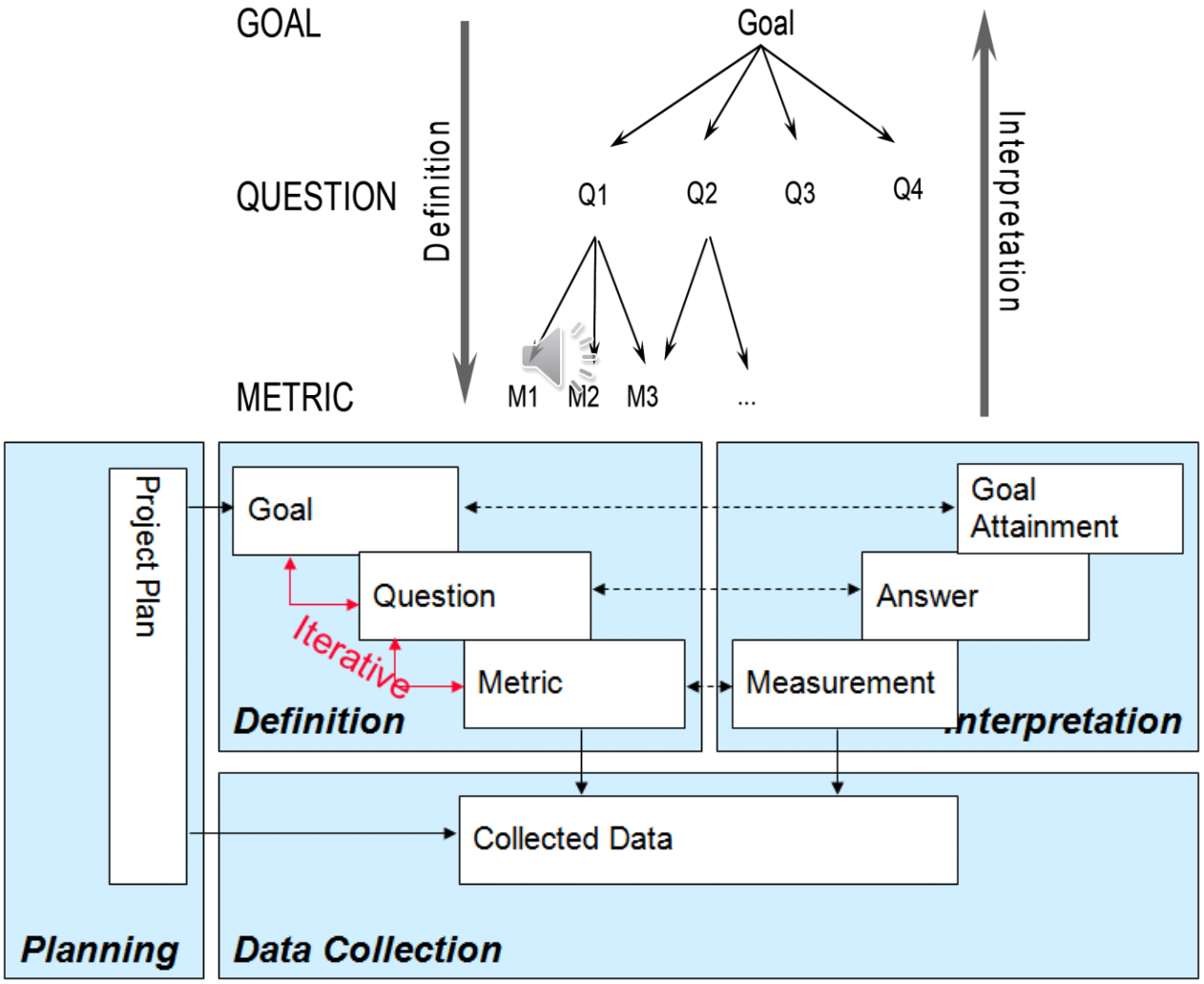
- The Experience Factory is a concept where an organization is divided into a project organization that executes a project and into an experience organization that collects and analyzes information


Project Organization
1 Characterize
2 Set Goals
3 Choose Process
Execution Plans
4 Execute
Environment Characteristics
Tailorable Knowledge
Products, Lessons Learned, Models
Project Analysis, Process Modification
Data, Lessons Learned
Experience Organization
Project Support
Experience Base
5 Analyze
6 Package
Generalize
Tailor
Formalize
Disseminate
Experience Factory

Process Improvement and Measurement - GQM

- Goal Question Metric (GQM) is a top-down method for defining measurements according to stated measurement goals
- Develop a set of measurement goals for aspects of the software product and process of interest (e.g., productivity and quality)
- Generate questions (based upon models) that define those goals as completely as possible in a quantifiable way
- Specify the measures needed to be collected to answer those questions and track process and product conformance to the goals
- Develop mechanisms for data collection
- Collect, validate, and analyze the data in real time to provide feedback to projects for corrective action
- Package knowledge based on the interpretations and make recommendations for future improvements

GQM specific measurement process



[van Solingen & Berghout, 1999]

Aligning Improvement Goals with Strategies

- Model-based and continuous improvement approaches can influence organizational activities
- However, chances are that the improvements are not associated with the business goals of the organization or that decision makers don't acknowledge the benefits of the improvement programs
- There is a need to show that a possible relationship exists between the organizational strategic objectives and the improvements made
- Measurement can help to connect strategic objectives with improvements but instrumentation and interpreting the results are not easy to do
- Concepts which specifically highlight measurement and strategic objectives include such as the Balanced Scorecard (BSC) and GQM^{+} Strategies

Balanced Scorecard (BSC)

- Balanced Scorecard (BSC) is an approach for measuring whether the activities of a company are meeting its goals with respect to vision and strategy
- BSC was initially developed by Robert S. Kaplan and David P. Norton (published 1992)
- BSC is used by decision makers for controlling organization-wide goals making use of different dimensions that have to be determined based on goals and characteristics of an organization
- In order to derive goals and document relationships among them, a strategy map is defined, which connects all dimensions by using causal links (aka. BSC stories)
- Typically, four different dimensions are controlled (learning & growth, internal process, customer, and financial) each including 1 to 2 business goals
- For each goal of a BSC dimension, corresponding indicators are defined, target values, and initiatives (activities)

BSC – Basic Model and Relationships

